

25PC2RA202: Sensors and Actuators Laboratory (S&A Lab)

Experiment 2: Inertial Data Analysis

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Student Name: _____

Number: _____

1. Introduction

In this experiment, you will measure noise and bias from an Inertial Measurement Unit (IMU) stream from the raw IMU data.

2. Hardware

- Arduino R4 WiFi Board
- USB C-to-C Cable
- 7SEMI 9-Axis IMU

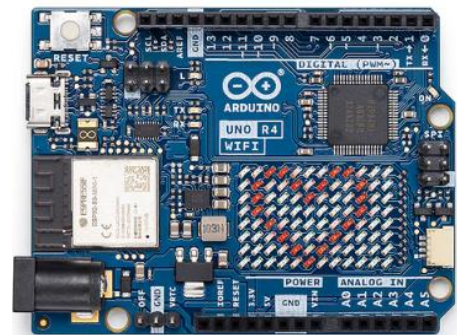


3. Wiring

IMU Pin	Signal	Arduino R4 Pin	Wire
3.3V	Supply Voltage	3.3V	Red
GND	Ground	GND	Black
SDA	Serial Data	SDA	Green
SCL	Serial Clock	SCL	Yellow/Blue



4. Arduino Code



```

1  /* =====
2  * EXPERIMENT: IMU bias & noise characterization
3  * Board: Arduino UNO R4 WiFi
4  * Sensor: 7Semi BNO055
5  *
6  * GOAL: Keep the sensor perfectly still for ~60 s and compute, for all
7  * six channels (ax ay az gx gy gz):
8  *     1. Bias = mean - expected value
9  *     2. Noise = sqrt( mean of squares - square of mean )
10 *
11 * TODO: Complete 1 to 4 in order. Do not modify setup() or loop().|
12 * ===== */
13 #include <Wire.h>
14 #include <7Semi_BNO055.h>
15
16 BNO055_7Semi imu;
17
18 const int N = 5000; // ~60 s of samples
19 const char* name[6] = {"ax", "ay", "az", "gx", "gy", "gz"};
20 const double expected[6] = {0, 0, 9.81, 0, 0, 0}; // True values (NO MOVEMENT OF THE SENSOR)
21
22 double sum[6] = {0, 0, 0, 0, 0, 0},
23 double sumSq[6] = {0, 0, 0, 0, 0, 0};
24 int n = 0;
25
26 // DO NOT TOUCH THIS FUNCTION
27 void setup() {
28     Serial.begin(115200);
29     while (!Serial);
30
31     if (!imu.begin())
32     {
33         Serial.println("BNO055 not found");
34         while (1);
35     }
36
37     imu.setMode(Mode::ACCGYRO); // Raw mode
38     delay(1000); // 1 second delay
39     Serial.println("Measuring for ~60 s - DO NOT TOUCH/MOVE THE SENSOR");
40 }
41
42 // DO NOT TOUCH THIS FUNCTION
43 void loop() {
44     int16_t axr, ayr, azr, gxr, gyr, gzh;
45     double s[6];
46
47     // TODO 1
48     if (!readSensors(axr, ayr, azr, gxr, gyr, gzh)) {
49         Serial.println("LINK LOST");
50         delay(10);
51         return;
52     }
53
54     // TODO 2
55     convertToUnits(axr, ayr, azr, gxr, gyr, gzh, s);
56
57     // TODO 3
58     accumulate(s);
59
60     if (n == N) {
61         // TODO 4
62         printResults();
63         while (1);
64     }
65     delay(10);
66 }

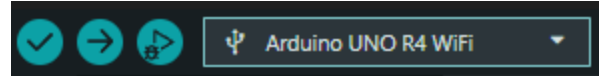
```

```

67
68  ✓ /* -----
69   * TODO 1 – Complete the following function: readSensors
70   * Ask the BNO055 for its raw accelerometer and gyroscope counts - Call
71   * imu.readAccel(axr, ayr, azr) and imu.readGyro(gxr, gyr, gzyr).
72   * ----- */
73  ✓ bool readSensors(int16_t &axr, int16_t &ayr, int16_t &azr,
74    || || || || || || int16_t &gxr, int16_t &gyr, int16_t &gzyr) {
75    // Your code here
76
77    return false; // <-- replace "false" with your result
78  }
79
80  ✓ /* -----
81   * TODO 2 – Complete the following function: convertToUnits (units)
82   * The sensor provides raw integer values.
83   * STEP 1: Convert:
84   *   s[0..2] = axr, ayr, azr divided by 100.0 → m/s2 (100 counts per m/s2)
85   *   s[3..5] = gxr, gyr, gzyr divided by 16.0 → °/s (16 counts per °/s)
86   * STEP 2: Pack into s[]
87   * ----- */
88  ✓ void convertToUnits(int16_t axr, int16_t ayr, int16_t azr,
89    || || || || || || int16_t gxr, int16_t gyr, int16_t gzyr, double s[6]) {
90    // Your code here
91
92  }
93
94  ✓ /* -----
95   * TODO 3 – Complete the following function: accumulate
96   * The mean needs only the running TOTAL;
97   * The standard deviation needs only the running TOTAL OF SQUARES.
98   * LOGIC:
99   * For each channel i = 0 to 5:
100   *   a) add s[i] into sum[i]
101   *   b) add s[i]*s[i] into sumSq[i]
102   * Then count this sample: increase n by 1.
103   * ----- */
104  ✓ void accumulate(double s[6]) {
105    // Your code here
106  }
107
108  ✓ /* -----
109   * TODO 4 – Complete the following function: printResults
110   * For each channel i = 0 to 5, compute and print:
111   *   mean = sum[i] / n
112   *   sigma = sqrt( sumSq[i]/n - mean*mean )
113   *   bias = mean - expected[i]
114   * Print one line per channel: name, bias, noise (use 4 decimals).
115   * ----- */
116  ✓ void printResults() {
117    Serial.println("---- RESULTS for Table 1 ---");
118    // Your code here
119
120  }

```

5. Procedure



- Complete the wiring as per Section 3 with the Arduino R4 disconnected from the computer.
- Connect the Arduino R4 to the computer.
- Place the IMU flat on the desk, facing up, and keep it still until the end of the experiment.
- Instructions on how to run arduino
 - Open Terminal (Ctrl + Alt + T)
 - Run this code: `./Applications/arduino-ide_2.3.10_Linux_64bit.Applmage`
 - Ensure selected board - Arduino Uno R4 WiFi
- Complete the **TODO (1/2/3/4)** sections in Arduino code. Verify & Upload the code.
- Open Serial Monitor. Change baud rate to 115200.
- Complete the observation table.

6. Observation Table:

Channel	Expected	Mean	Bias	Noise
Acceleration a_x (in m/s^2)				
Acceleration a_y (in m/s^2)				
Acceleration a_z (in m/s^2)				
Angular Velocity ω_x (in $^\circ/s$)				
Angular Velocity ω_y (in $^\circ/s$)				
Angular Velocity ω_z (in $^\circ/s$)				

7. Evaluation Rubric

Assessment Component	Marks	Awarded Marks
Wiring	1	
Programming	5	
Observation Table	3	
Attendance	1	
Total	10	

Signatures

Prof. A. Adithya Kashyap
Date:

Prof. Harshvardhan Uppaluru
Date: